Research Statement

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Ph.D. in Mathematics

1. Research Vision

My research program follows the chain:

$$\Big(\text{MATHEMATICS} \xrightarrow{\text{Explain}} \text{AI SYSTEMS}\Big) \xrightarrow{\text{Enable}} \text{RELIABLE ALIGNMENT}$$

where mathematics explains how AI systems learn, and AI systems, once understood, become more reliable.

The first link is mathematical. Modern AI systems succeed across a remarkable range of domains, yet a principled account of why they succeed remains incomplete. My dissertation works on this question by treating the weight Gram matrix of a neural network as the central object of analysis. From a single chain-rule identity, it follows that gradient descent shifts this object in a way that progressively, layer by layer, unpacks a nonlinear data manifold into linear representations. The next question my proposed research takes up is what this implicit regularization implies for generalization.

The second link is more practical. AI systems are increasingly adopted in real life, yet most alignment methods operate end-to-end: a model is trained to produce the desired outputs, and what changes inside it is never observed. These methods optimize a single scalar loss and hope the model's internals align as intended, yet the resulting alignment is fragile: a single direction in activation space can undo it [2]. I have worked on AI alignment problems, from hallucination to AI cooperation, and my current work extends this theoretical understanding into a transparent and backpropagation-free alignment algorithm.

2. Past & Current Research

Past Research

Mathematical Foundations of AI A common thread runs through my mathematical work on AI: the geometry of the data space is the right object of analysis. **ABC3** [5] addresses sample-efficient causal inference in randomized experiments. I recast the imbalance between treatment and control groups as a *distance between their distributions in a kernel-induced feature space*. **Curse of Smoothness** [6] identifies a failure mode in functional neural networks, which enforce smoothness on hidden states and weight functions. I prove that *this functional smoothness bounds the weight gradient norm by a product of Lipschitz constants that compounds across depth*, producing a vanishing-gradient effect that ReLU activations cannot rescue.

Alignment in Practice, and Its Limits Another thread I have been focusing on during my PhD is quantifying and improving the reliability of modern AI systems. I began with a striking observation: pre-trained language models return a **distinctive language modeling distribution for hallucinated texts** [4]. Through extensive analysis, I proposed a simple hallucination mitigation method for generative language models. The **Concordia Contest at NeurIPS 2024** [17] asked participants to build agents that cooperate with other AI agents. Building on the Common Ground concept, learned through sociological training, I won first place out of 197 participants.

Carrying out this work across theory and practice, I found a significant gap between them: without a proper understanding of modern AI systems, we are left with post hoc benchmarks and no account of why they pass or fail. My earlier work suggests that geometry is the right level of analysis for AI systems. Closing the gap therefore requires knowing the geometric mechanism by which deep networks reshape data in the first place: what is the geometry of feature learning?

Current Research

In my dissertation and the resulting preprint [7], accepted for NeurIPS 2026, joint work with Adityanarayanan Radhakrishnan (MIT) and Daniel Beaglehole (UCSD), I derived the following exact identity. For any function f of an input $z = Wh$,

$$W^\top \nabla_W f = \nabla_h f \cdot h^\top, \tag{1}$$

which follows from the chain rule alone, with no assumption on architecture, loss, or data. I call this the *feature learning equation*, as it bridges the gradients in weight space and in feature space. Two facts follow from it.

Exact training dynamics beyond the Neural Feature Ansatz. The Neural Feature Ansatz [15] conjectures that the weight Gram matrix satisfies $W^\top W \propto \mathbb{E}\left[\nabla_h f \cdot \nabla_h f^\top\right]$. This relationship has been central to explaining a range of phenomena in neural networks, including Neural Collapse [3] and grokking [11]. The ansatz, however, has been established only under restrictive assumptions [16], and settings are known in which it fails [18].

In the dissertation, I show that for one gradient step $W^+ = W - \gamma \nabla_W \mathscr{L}$, substituting Equation (1) gives

$$(W^+)^\top W^+ - W^\top W = h^+ (h^+)^\top - hh^\top + O(\gamma^2), \qquad h^+ := h - \gamma \nabla_h \mathscr{L}. \tag{2}$$

Here h^+ is the *virtual update*: the state gradient descent would produce if features, rather than weights, were the optimization variable. No such update is ever applied, yet $W^\top W$ tracks the hidden state's *virtual* covariance structure.

Training and layerwise dynamics of implicit linearization. Implicit linearization in deep networks is known under several names, including linear probing [1], the law of data separation [9], and Neural Collapse [14]. Theoretical understanding beyond idealized settings remains limited, since existing accounts rely on strong assumptions such as the unconstrained features model [13], in which hidden features are directly optimizable.

For hidden states $H \in \mathbb{R}^{d \times N}$ and targets Y, I define *Target Linearity* (TL) as the R^2 of the ridge regressor $\hat{Y}_\lambda = H^\top (HH^\top + \lambda I)^{-1} HY$. TL is defined at every layer and every training step, for any task with a target. I derive the training and layerwise dynamics of this quantity and show that it grows with depth in interpolating networks, so *a deep network sequentially unpacks nonlinear data into progressively more linear representations*. Two known phenomena follow as special cases: Neural Collapse is the extreme of target linearization, and high TL in a latent space is sufficient for linear interpolation in generative models.

3. Proposed Research

Thrust 1. A generalization theory from implicit linearization dynamics Classical theory bounds a deep network's complexity globally, and the resulting bounds are loose: the same theory also permits memorization of random labels [19], and no complexity measure predicts generalization reliably [10]. Sequential linearization offers a way around this. Under this view each layer has an implicit objective, to return a hidden state from which the target is more linearly recoverable than from the one below, and the work a layer performs is the TL increment it produces. If one layer does nearly all the linearization work, it implements a large transformation and its own generalization error is correspondingly large; if the linearization work is divided, each transformation is small. Depth then helps not only by adding expressivity but also by letting the required nonlinearity be distributed.

> **Conjecture 1** *Generalization is determined by how evenly the work of linearization is distributed across depth. Networks in which every layer contributes a comparable* TL *increment generalize; networks in which the increment concentrates in a few layers memorize.*

This reframing makes classical linear model theory available: bounds for ridge regression are sharp and rarely vacuous. The work proceeds in two steps. First, I will assign each layer a complexity term by treating it as a regression from the features below, measured by the weight norm and feature radius that the surrogate $Y^\top H^\top W^\top WHY$ already controls, so that a larger TL increment carries a larger term. This turns a trained network into a vector of L numbers rather than a single complexity score. Second, I will define a uniformity metric on that vector and prove that the network's generalization gap is controlled by it, so that a peaked profile gives a worse guarantee than a flat one at equal total work.

Preliminary Results and Supporting Work: In grokking on modular addition, the gap between the middle and last layers narrows sharply at the onset of generalization, as the middle layer takes up a share of the linearization it had not been performing before. The transition is visible in the increment profile, not only in the final TL value. Independent theoretical support comes from the observation that gradient flow on homogeneous networks keeps the norm gap between adjacent layers bounded throughout training, so layers are automatically balanced [8]. That result concerns the balance of weight norms; whether it extends to a balance of linearization work is what Conjecture 1 asks.

Thrust 2. Understanding linear representations for reliable alignment method The linear representation hypothesis, first motivated by "King $-$ Man $+$ Woman $\approx$ Queen" [12], underlies several alignment methods [20], yet we do not know why it holds or when it fails. Linearization offers an explanation. If a network linearizes its input manifolds with respect to the same target, two manifolds carrying distinct concepts are pushed toward the same linear structure, so an affine map between them should exist.

Conjecture 2 *When a network is trained to linearize input manifolds with respect to a target task, there exists an affine transformation mapping the hidden representation of one manifold onto that of the other, where corresponding points are defined by shared task structure.*

Concept Gradient: transparent alignment from model internals. This theory grounds a new alignment method. Existing approaches sit at two extremes: fine-tuning and RLHF backpropagate through scalar loss, while activation steering works inside the model but acts at inference time in a single shift. Equation (1) gives a third route. For a concept vector s from contrastive texts, setting $\nabla_h f := s$ yields $\Delta W = (W^\top)^\dagger s \cdot h_{\text{cal}}^\top$, where $\dagger$ is the pseudoinverse and h_{cal} are calibration activations. Because the update enters the weights, it can be applied repeatedly, controlling behavior directly through the model's internals I call this backpropagation-free method the *concept gradient*.

Preliminary Results: On Llama3 8B, this update shifts the targeted behavior monotonically over several steps while leaving perplexity and untargeted behavior at baseline, where a comparable single activation shift degrades both. Because calibration activations enter as an outer product, pairing a concept vector with a chosen calibration set installs a conditional association rather than a uniform shift, letting two opposite associations coexist in one model. More strikingly, when the concept gradient is used to impose the simplex ETF structure of Neural Collapse on the feature space of a randomly initialized network, the resulting training and generalization curves closely track those of SGD. This indicates that the concept gradient is not only a practical control tool, but can also serve as a feature-space dual of gradient descent through which neural network training can be analyzed.

4. Long-term Vision

My ultimate goal is to understand how the simplest linear arguments can explain phenomena emerging in complicated modern AI systems. By explaining two prominent phenomena — generalization and linear representations — I believe the Target Linearity concept can fundamentally change how we think about deep neural networks: though they look inevitably complex, they can be decomposed into much simpler submodules. Because this same linear structure is what makes concepts identifiable and steerable, I see it as the mathematical bridge my research vision promised at the outset: a theory of how networks represent the world is, at the same time, a theory of how to reliably control what they represent.

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